Minah Oh

CONTACT INFORMATION

Department of Mathematics and Statistics
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CITIZENSHIP

United States Citizen

EDUCATION

Ph.D. in Mathematics, University of Florida, Gainesville, FL — May 2010
Dissertation Title: Efficient Solution Techniques for Axisymmetric Problems
Advisor: Dr. Jay Gopalakrishnan
MS in Mathematics, University of Florida, Gainesville, FL — May 2007
BS in Mathematics, Yonsei University, Seoul, Korea — Aug. 2005
Exchange Student, St. Olaf College, Northfield, MN — 2003–2004

POSITIONS

Professor, Department of Mathematics and Statistics — Aug. 2022–present
James Madison University, Harrisonburg, VA
Associate Professor, Department of Mathematics and Statistics — Aug. 2016–May 2022
James Madison University, Harrisonburg, VA
Assistant Professor, Department of Mathematics and Statistics — Aug. 2010–May 2016
James Madison University, Harrisonburg, VA
Graduate Teaching Assistant — Aug. 2005–May 2010
University of Florida, Gainesville, FL
Intern, Citigroup, Sales and Trading in Emerging Markets, Seoul, Korea — 2005
Intern, Tong Yang Venture Capital, Seoul, Korea — 2003

RECENT POSITIONS AND AWARDS

Mathematical Association of America (MAA)
Congress Representative of the Maryland-DC-Virginia Section — 2026–2030

2025 MAA Meritorious Service Award — Aug. 2025

2022 Sister Helen Christensen Service Award, MAA MD-DC-VA Section — Nov. 2022

RESEARCH INTERESTS

Numerical Partial Differential Equations: Finite Element Methods and Finite Element Exterior Calculus, Efficient Solution Techniques for Axisymmetric Problems and their Applications, Optimal Control Problems

Numerical Linear Algebra: Low-Rank Approximation of Matrices and its Applications in Data Science

PUBLICATIONS

- Research Papers
  - M. Oh, L. Ma, and K. Wang, *P_1 finite element methods for a weighted elliptic state-constrained optimal control problem*, Numer. Algorithms, 87 (2021), pp. 1–17, DOI.
  - M. Oh, *The Hodge Laplacian on axisymmetric domains and its discretization*, IMA J. Numer. Anal., 41 (2021), pp. 1569–1607, DOI.
  - M. Oh, *Multigrid in $H(\mathrm{div})$ on axisymmetric domains*, J. Math. Anal. Appl., 490 (2020), 124209, DOI.

- S. C. Brenner, M. Oh, and L.-Y. Sung, *P_1 finite element methods for an elliptic state-constrained distributed optimal control problem with Neumann boundary conditions*, Results Appl. Math., 8 (2020), 100090, DOI.

- S. C. Brenner, M. Oh, S. Pollock, K. Porwal, M. Schedensack, and N. S. Sharma, *A C^0 interior penalty method for elliptic distributed optimal control problems in three dimensions with pointwise state constraints*, in Topics in Numerical Partial Differential Equations and Scientific Computing, S. C. Brenner, ed., IMA Vol. Math. Appl. 160, Springer, New York, 2016, pp. 1–22, DOI.

- M. Oh, *de Rham complexes arising from Fourier finite element methods in axisymmetric domains*, Comput. Math. Appl., 70 (2015), pp. 2063–2073, DOI.

- M. Oh, *Introducing proofs to calculus students*, in Beyond Lecture: Resources and Pedagogical Techniques for Enhancing the Teaching of Proof-Writing Across the Curriculum, R. Schwell, A. Steurer, and J. F. Vasquez, eds., MAA Notes 85, Mathematical Association of America, Washington, DC, 2016, pp. 283–290.

- S. Bittner, J. Ducey, X. Guo, M. Oh, and A. Zweber, *Integer invariants of an incidence matrix related to Rota's basis conjecture*, Missouri J. Math. Sci., 29 (2017), pp. 27–32, DOI.

- M. Oh, *A new approach to the analysis of axisymmetric problems*, IMA J. Numer. Anal., 34 (2014), pp. 1686–1700, DOI.

- J. Gopalakrishnan and M. Oh, *Commuting smoothed projectors in weighted norms with an application to axisymmetric Maxwell equations*, J. Sci. Comput., 51 (2012), pp. 394–420, DOI.

- D. Copeland, J. Gopalakrishnan, and M. Oh, *Multigrid in a weighted space arising from axisymmetric electromagnetics*, Math. Comp., 79 (2010), pp. 2033–2058, DOI.

- Chen, Kong, Oh, Sanan, and Wang, *Visual words, text analysis concepts for computer vision*, IMA Mathematical Modeling in Industry XIII report, Institute for Mathematics and its Applications, Minneapolis, MN, 2009. Mentor: B. Wohlberg.

- Dissertation

  - M. Oh, *Efficient Solution Techniques for Axisymmetric Problems*, Ph.D. thesis, University of Florida, Gainesville, FL, 2010, Link.

UNDERGRADUATE RESEARCH STUDENTS PUBLICATIONS

- N. Stock, *Higher order Fourier finite element methods for Hodge Laplacian problems on axisymmetric domains*, SIAM Undergrad. Res. Online, 14 (2021), DOI. Advisor: M. Oh.

- K. Keegan, D. Melendez, and J. Zheng, *Media processing and a modified watermarking scheme based on the singular value decomposition*, SIAM Undergrad. Res. Online, 14 (2021), pp. 446–467, DOI. Advisor: M. Oh.

RESEARCH GRANTS

National Science Foundation, grant number DMS-1913050, Jul. 2019–Jun. 2023, "RUI: Efficient Numerical Methods for Axisymmetric Problems", $100,000 Awarded.
More information about my NSF-funded project can be found here.

JMU College of Science and Mathematics Summer Research Grant 2015 and 2011
$4,000

UNDERGRADUATE RESEARCH STUDENTS

Faculty Advisor for summer@ICERM2020 Fast Learning Algorithms for Numerical Computation and Data Analysis, Brown University Summer 2020

1. Katherine Keegan, David Melendez, and Jennifer Zheng
   Project Title: Randomized Singular Value Decomposition and its Applications
   Final Report
   ⋆ Publications in SIAM Undergraduate Research Online and SIAM Review

2. Trevor Crupi, Yonah Moise, and Hannah Odom
   Project Title: Multigrid for Fourier Finite Element Methods on Axisymmetric Domains
   Final Report

Andre Mas, JMU 2022–2023
Project Title: Fourier Finite Element Methods for Elliptic State Constrained Optimal Control Problems on Axisymmetric Domains with General Data
⋆ Presentation First Prize, MAA MD-DC-VA Section Meeting, Spring 2023

Nicole Stock, JMU 2020–2021
Honors Thesis Title: Higher Order Finite Element Methods for Axisymmetric Hodge Laplacian Problems
⋆ Poster Presentation First Prize, MAA MD-DC-VA Section Meeting, Spring 2021
⋆ Publication in SIAM Undergraduate Research Online

Grant Henderson and Jacob Spangler, JMU Spring 2020
Project Title: Fourier-FEMs for Elliptic State-Constrained Optimal Control Problems on Axisymmetric Domains

Brendan Armani, Andrew Levy, and Andrew Tomassone, JMU Spring 2018
Project Title: Analysis of the Singular Value Decomposition Applied to Image Compression
⋆ Student Presentation Second Prize, MAA MD-DC-VA Section Meeting, Spring 2018

Sophia Mancini and Unyoung Park, JMU Independent Studies Spring 2017
Project Title: The Singular Value Decomposition and its Applications
⋆ Poster Presentation First Prize, MAA MD-DC-VA Section Meeting, Spring 2017

Ben Rhodes, JMU Independent Studies Spring 2017–Spring 2018
Project Title: Numerical Methods for Axisymmetric Problems and deal.II

Charles Crook, JMU Internal REU Summer 2013
Project Title: Finite Element Methods for the Poisson Equation and its Applications

Justin Hall, JMU Internal REU Summer 2013
Project Title: Finite Element Methods for the Axisymmetric Maxwell Equations

Jacob Rhodes, JMU Independent Studies 2012–2013
Project Title: Finite Element Methods and 3D Printing

NSF REU (Co-mentored with Josh Ducey of JMU) Summer 2012
Students: Stephanie Bittner, Michael Cheung, Xuyi Guo, and Adam Zweber
Project Title: Approaches to Rota's Basis Conjecture

COURSES TAUGHT

Numerical Linear Algebra
Numerical Partial Differential Equations
Partial Differential Equations and Fourier Series
Computers and Numerical Algorithms
Differential Equations
Linear Algebra
Linear Algebra with Differential Equations
Methods of Applied Calculus
Calculus 1, 2, and 3
Business Calculus
Nature of Mathematics (General Education Course)
Special Topics Course: Finite Element Methods

PRESENTATIONS

- Conference Presentations

  – **Research Presentations**

AWM Research Symposium, University of Wisconsin-Madison May 2025
Quad-Curl Problems on Axisymmetric Domains

AMS Spring Southeastern Sectional Meeting, Georgia Institute of Technology, Special Session on Recent Development in Advanced Numerical Methods for Partial Differential Equations Mar. 2023
Multigrid for Axisymmetric H(curl)-Problems and its Applications

Finite Element Circus, Carnegie Mellon University Oct. 2022
Multigrid for Axisymmetric H(curl)–Problems

Afternoon Invited Address
Fall 2021 MAA MD-DC-VA Sectional Meeting Nov. 2021
Amazing Applications of Linear Algebra

50th Anniversary of the Finite Element Circus Nov. 2020
P1 Finite Element Methods for a Weighted Elliptic State-Constrained Optimal Control Problem

AMS Fall Central Sectional Meeting, Special Session on Recent Advances in Scientific Computing and Applications Sep. 2020
Multigrid in H(div) on Axisymmetric Domains

16th Copper Mountain Conference On Iterative Methods, Multigrid in H(div) on Axisymmetric Domains (Canceled due to COVID-19.) Mar. 2020

Joint Mathematics Meeting, Denver, CO Jan. 2020
P1 Finite Element Methods for Elliptic Optimal Control Problems

Finite Element Circus, Virginia Tech. Nov. 2019
Elliptic State-Constrained Optimal Control Problems with Neumann Boundary Conditions

MAA MD-DC-VA Section Meeting, Hood College & Frederick County Community College Apr. 2019
Using Mathematics to Solve Real-World Problems

Joint Mathematics Meeting, Baltimore, MD Jan. 2019
The Hodge Laplacian on Axisymmetric Domains

Finite Element Circus, University of Delaware Nov. 2018
The Hodge Laplacian on Axisymmetric Domains

MAA MD-DC-VA Sectional Meeting, CNU Nov. 2017
Grad, Curl, and Div on Axisymmetric Domains

Cascade RAIN Meeting, Washington State University Apr. 2016
Tools to Analyze Axisymmetric Problems

Finite Element Circus, George Mason University Mar. 2015
De Rham Complexes arising from Fourier-FEMs in Axisymmetric Domains

Finite Element Circus/Rodeo, Louisiana State University Mar. 2013
A New Approach to the Analysis of Axisymmetric Problems

2013 Joint Mathematics Meeting, San Diego, CA Jan. 2013
A New Approach to the Analysis of Axisymmetric Problems

2011 Joint Mathematics Meeting, New Orleans, LA Jan. 2011
Commuting Smoothed Projections in Weighted Spaces

Finite Element Circus, IMA, Minneapolis, MN Nov. 2010
Commuting Smoothed Projections in Weighted Spaces

33rd SIAM Southeastern-Atlantic Section Conference Apr. 2009
University of South Carolina
Recent Advances in Computational Electromagnetics

– **Presentations related to Undergraduate Education**

2026 MathFest, Boston, MA Aug. 2026
Spreading the Love of Mathematics

MAA MD-DC-VA Section Meeting, Frederick, MD Apr. 2026
Spreading the Love of Mathematics

Joint Mathematics Meeting, Denver, CO Jan. 2020
How Exposure to Research in Numerical PDEs Influenced Students In My PDE Class

Joint Mathematics Meeting, Baltimore, MD Jan. 2019
Teaching Linear Algebra through its Applications

MAA MD-DC-VA Section Meeting, UMW — Nov. 2018
The Value of Solid Mathematics for Computer Algorithms

MAA MD-DC-VA Sectional Meeting, Johns Hopkins University — Nov. 2016
Effective In-Class Programming Projects for STEM Majors

JMU Kappa Delta Pi's Annual Educators' Conference — Feb. 2016
Making a Difference with Teaching

MAA MD-DC-VA Sectional Meeting, Roanoke College — Mar. 2015
Finite Element Methods and Undergraduate Research

MAA MD-DC-VA Section Meeting, Harrisonburg, VA — Apr. 2014
Introducing Proofs to Calculus Students

2013 Joint Mathematics Meeting, San Diego, CA — Jan. 2013
An Applied Project for Linear Algebra Students: Finite Element Methods

MAA MD-DC-VA Section Meeting, Newport News, VA — Nov. 2011
An Applied Project for Linear Algebra Students

2011 Joint Mathematics Meeting, New Orleans, LA — Jan. 2011
Introducing Proofs to Calculus Students

- Invited Colloquium and Seminar Talks

Colloquium, Salisbury University, Virtual Colloquium, — Apr. 2021
Colloquium, Colorado School of Mines, Virtual Colloquium, — Oct. 2020
Colloquium, University of North Florida, Jacksonville, FL — Feb. 2019
Colloquium, St. Olaf College, Northfield, MN — Oct. 2018
Numerical Analysis Seminar, Louisiana State University — Oct. 2018
Numerical Analysis Seminar, Mississippi State University — Sep. 2018
Colloquium, University of Mary Washington — Apr. 2016
Applied & Computational Mathematics Seminar
Portland State University, Portland, OR — Apr. 2016
Colloquium, Longwood University, Farmville, VA — Sep. 2013
Numerical Analysis Seminar, Virginia Tech, Blacksburg, VA — Nov. 2011
Colloquium, Colgate University, Hamilton, NY — Feb. 2010
Colloquium, University of North Carolina at Asheville — Feb. 2010
Colloquium, Clarkson University, Potsdam, NY — Feb. 2010
Colloquium, James Madison University, Harrisonburg, VA — Jan. 2010

WORKSHOPS

(Invited) Mathematisches Forschungsinstitut Oberwolfach Workshop — Jun. 2022
Hilbert Complexes: Analysis, Applications, and Discretizations
Oberwolfach, Germany. (Institute offering board and lodging.)

Workshop at East Coast Optimization Meeting (ECOM) 2021 — Apr. 2021
Optimization for Machine Learning
Virtual Conference

Workshop at CM2021 — Mar. 2021
20th Copper Mountain Conference on Multigrid Methods
Algebraic Multigrid and Advanced Topics, Parallel Multigrid
Virtual Conference

ICERM Semester Workshop Jun. 2018
Computational Aspects of Time Dependent Electromagnetic Wave Problems in Complex Materials
ICERM, Providence, RI. (Attended with full travel support by organizers.)

ICERM Semester Workshop Oct. 2017
Mathematical and Computational Aspects of Radar Imaging
ICERM, Providence, RI.

IMA Special Workshop Mar. 2015
WhAM! A Research Collaboration Workshop for Women in Applied Mathematics: Numerical Partial Differential Equations and Scientific Computing
Louisiana State University. (Attended with full travel support by organizers.)

IMA Special Workshop Oct. 2014
Structure-Preserving Discretizations of Partial Differential Equations
IMA, Minneapolis, MN. (Attended with full travel support by organizers.)

IMA Special Workshop Aug. 2014
WhAM! A Research Collaboration Workshop for Women in Applied Mathematics: Numerical Partial Differential Equations and Scientific Computing
IMA, Minneapolis, MN. (Attended with full travel support by organizers.)

NSF/CBMS Conference Jun. 2012
Finite Element Exterior Calculus (FEEC)
ICERM, Providence, RI. (Attended with full travel support by organizers.)

IMA Annual Program Year Workshop Jun. 2011
Large-scale Inverse Problems and Quantification of Uncertainty
IMA, Minneapolis, MN.

IMA Annual Program Year Workshop Nov. 2010
Numerical Solutions of Partial Differential Equations: Fast Solution Techniques
IMA, Minneapolis, MN. (Attended with full travel support by organizers.)

IMA Math Modeling Mathematical modeling in industry XIII Aug. 2009
A Workshop for Graduate Students
IMA, Minneapolis, MN. (Attended with full travel support by organizers.)

HONORS AND AWARDS

Meritorious Service Award, MAA Aug. 2025

Sister Helen Christensen Service Award, MAA MD-DC-VA Section Nov. 2022

Section Project NExT 2010–2012

Graduate Teaching Award, University of Florida Mar. 2007
20 awards given among several thousands of Teaching Assistants in the university.

Departmental Teaching Award Mar. 2007
Department of Mathematics, University of Florida

SIAM Student Chapter Certificate of Recognition May 2008
The SIAM Education Committee

PROFESSIONAL ASSOCIATION

Mathematical Association of America (MAA)
American Mathematical Society (AMS)
Society for Industrial and Applied Mathematics (SIAM)
Association for Women in Mathematics (AWM)

PROFESSIONAL SERVICE

- **Professional Societies or Grant Agencies**

MAA Congress Representative for the MD-DC-VA Section	2026–2030
Past Chair, MD-DC-VA Section of the MAA	2021–2022
Chair, MD-DC-VA Section of the MAA	2019–2021
Program Chair, MD-DC-VA Section of the MAA	2016–2018
NSF panelist	2019, 2020, 2022, 2025, 2026
Conference Organizer MAA MD-DC-VA Virtual Sectional Meeting	Apr. 2021 and Nov. 2020
Project NExT Consultant	2020–present
MAA Tensor Women and Mathematics Grant Panelist	2015 and 2017–2019
MAA Committee on the Participation of Women (CPW)	2018–2024
Chair Elect, MD-DC-VA Section of the MAA	2018–2019
MAA MD-DC-VA Section's Liaison with the AWM	2016–present
At Large Member MD-DC-VA MAA Section Meeting	2013–2015
Paper Referee SIAM Journal on Numerical Analysis (SINUM), Computers and Mathematics with Applications (CAMWA), Journal of Scientific Computing (JSC), Journal of Computational Physics (JCP), Results in Applied Mathematics (RINAM), and Computer Methods in Applied Mechanics and Engineering (CMAME)	2011–present
Regular Undergraduate Student Poster and Paper Judge at Professional Meetings	

- **Department, College, and University**

Academic Program Review Committee	Aug. 2024–present
Chair of the Applied Math Committee	Aug. 2021–May 2024
Personnel Advisory Committee	Aug. 2019–May 2021
Member of JMU Asian Pacific Islander Desi Americans (APIDA) Caucus	Sep. 2021–present
JMU AWM Student Chapter Faculty Advisor	2012–2013, 2016–present

Established the JMU AWM Student Chapter. Dec. 2011

Applied Math Committee Fall 2010–present

Student Activities Committee Fall 2014–present

Academic Advisor for Math Majors Fall 2011–present

Academic Program Review Committee Spring 2015–Spring 2016

Student Awards Committee 2014–2015

Student Evaluations of Teaching Committee 2014–2015

Colloquium Committee 2013–2014

Search Committee 2012–2013

- **Outreach Activities**

Creator and Organizer of Girls' Math and Coding Day Mar. 2020–present
at JMU (an annual outreach event where 30 local female high school students from the AVID program come to James Madison University and learn mathematics and its applications with programming)

Workshop leader for madiSTEM Apr. 2011 and Apr. 2015
James Madison University (a one-day math and science conference for girls in grades 6–8 in the Shenandoah Valley)

Volunteer Math Tutor Summer 2010
The Arbor House (homeless shelter for single moms) Gainesville, FL

- **Others**

Conference Advisor 2008–2009
The Second SIAM Gators Student Conference, University of Florida

President, SIAM Student Chapter, University of Florida 2007–2008

SKILLS

Computer skills: Fluent in Python, MATLAB, and C/C++ Programming, Git, and LaTeX.
Language: Bilingual in English and Korean.